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RHEOLOGY OF THE VITREOUS BODY: PART 2. VISCOELASTICITY OF BOVINE AND PORCINE VITREOUS

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ABSTRACT The viscoelastic properties of the vitreous body from bovine and porcine eyes were determined by microrheometry. Each vitreous sample was sectioned into anterior, central and posterior segments. The rheological properties of each species and region were compared with each other and with the viscoelastic properties of human vitreous reported earlier. The results showed significant variations among species, as well as between regions in all species. All regions of human vitreous have significantly different rheological properties from those of the cow, and from the anterior and posterior regions of the pig; however, the viscoelastic behavior of the central porcine region closely resembles that of the human. In comparing the three species, the major differences found are in those parameters characterizing "viscous" or dissipative properties; values of "elastic" or energy storage properties were similar.

Introduction

The rheological behavior of human and animal eyes is being studied in an investigation of the viscoelastic properties of the vitreous body to provide insight into the determinants of vitreous gel properties, and to understand the changes that occur within the vitreous as a result of age and disease. In the first paper of this series (Lee *et al.*, 1992), a detailed description of the instrumentation and methods used for measuring creep compliance of the vitreous was presented. For ease of comparison, the compliance was analyzed using a four-parameter viscoelastic model, and results on human eyes were presented.

Although the vitreous of all species is basically composed of the same extracellular matrix elements, previous studies indicate that there are species variations in the concentration of the vitreous structural elements, principally hyaluronic acid (HA) and collagen (Balazs *et al.*; 1959; Balazs, 1984). A reasonable hypothesis is that these differences are responsible for the various physical states observed in different species. The vitreous body is not a gel in all species. In some, it is partly or completely liquid; in mammals, it is usually a gel in healthy adult eyes except in a few primates (Balazs, 1960). The

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variations in the physical state and consistency of the vitreous in different species may well be related to the animals' visual function, suggesting the need for a comparative analysis of vitreous structure and function. In this paper, we present data on the viscoelastic behavior of vitreous samples derived from bovine and porcine eyes, and compare these results with those previously reported on humans. In the following papers of the series, we will discuss studies of the chemical composition of the vitreous from these species, in an attempt to correlate and understand differences in rheological behavior of the vitreous between species. Ultimately, correlating this information with information on other features of different animal eyes and animal behavior, forms of activity and possibly environment, should lead to an in-depth understanding of vitreous structure and function.

Materials and Methods

Porcine and bovine eyes were readily available from local slaughterhouses, and could be obtained on a regular basis within 3–5 hours after the animals were slaughtered. Porcine eyes were given generously by Peter Villari Inc. (Philadelphia, PA). Bovine eyes were purchased from Fischera Meat Co. (Philadelphia, PA). Eyes were transported on ice, and refrigerated in a moist environment until dissection. Each eye was sectioned into Anterior, Central, and Posterior segments as described previously (Lee *et al.*, 1992).

All samples were studied rheologically using the microrheometer technique to determine creep compliance, and in most cases, the compliance was fit to a four-parameter model. In some samples, particularly from porcine eyes, the elasticity was such that a three-parameter model gave an adequate fit to the creep data. Two additional parameters were calculated from these, giving up to six rheological values for each sample which could be used for comparison and statistical analysis. These parameters are:

η_m unrecoverable (Maxwell) viscosity, dyne-sec/cm^2

η_k internal (Kelvin) viscosity, dyne-sec/cm^2

τ_m relaxation time, sec

τ_k retardation time, sec

J_m instantaneous (Maxwell) elastic compliance, cm^2/dyne

G_k internal (Kelvin) modulus, dyne/cm^2

As noted previously (Lee *et al.*, 1992), only four of these six parameters are independent, since a four-parameter model is fit to the data. Where the three-parameter model gave the best fit, the value of J_m is set to zero; this is not arbitrary, but based on the best data fit using both models, which indicate that the value of J_m is not significantly different from zero, so that the best values of the other parameters can be obtained with the three-parameter model. For comparative purposes, all six parameters are used here, since each has a different physicochemical interpretation that can aid in understanding the results. However, apparent correlations may appear because of functional relationships among the variables.

Statistical analyses were carried out using Multivariate Analysis of Variance (MANOVA) to evaluate the variability among regions and species, and Analysis of Variance (ANOVA) to determine group differences (SAS Institute, Inc., Cary, N.C.).

Results

Values of the six rheological parameters for bovine and porcine eyes are given in Tables 1 and 2 for each of the three regions. Each entry is the mean value plus or minus one standard deviation. The number in parentheses is the number of samples whose value for that particular parameter was within 2.5 standard deviations of the mean (Lee *et al.*, 1992). The relatively large standard deviations are a normal reflection of the individual variation of rheological properties from eye to eye, since they are significantly greater than the precision in determining the data, which is less than 10–15% for the parameters given (Lee *et al.*, 1992).

Table 1

Rheological Values for Bovine Vitreous as a Function of Region

	η_m	η_k	τ_m	τ_k	J_m	G_k
Anterior	144.52 +/- 74% (N=19)	10.07 +/- 75% (N=18)	9.27 +/- 80% (N=20)	0.98 +/- 81% (N=20)	0.06 +/- 50% (N=19)	11.60 +/- 55% (N=18)
Central	197.97 +/- 64% (N=18)	17.07 +/- 76% (N=19)	8.87 +/- 60% (N=19)	0.81 +/- 52% (N=19)	0.05 +/- 60% (N=20)	17.31 +/- 33% (N=17)
Posterior	255.31 +/- 61% (N=19)	21.07 +/- 81% (N=19)	10.47 +/- 53% (N=19)	0.69 +/- 54% (N=18)	0.05 +/- 60% (N=20)	26.93 +/- 51% (N=19)

Analysis of the data using both a t-test and Tukey's studentized test range (Lee *et al.*, 1992; Havilcek and Crain, 1988) for the bovine and porcine data allows determination of significant pairwise differences in rheological parameters between regions. For the bovine vitreous, G_k showed significant differences between all three regions; the two viscosity parameters, η_m and η_k, showed significant differences between the anterior and posterior regions only; and no differences between regions for the other three parameters. Behavior of the porcine samples was quite different: J_m is significantly different between all three regions, while G_k shows no differences at all between regions. The other four parameters show significant differences between the anterior and central regions, and between the central and posterior regions, but no differences between anterior and posterior.

In order to understand the significance of these results, and to facilitate comparison of the rheological parameters for the various regions and species, we present in Figs. 1–6 each of the parameters for the bovine and porcine vitreous from Tables 1 and 2, as well as for human vitreous from our earlier paper (Lee *et al.*, 1992). Each of these will be discussed in turn. Finally, the results of the detailed statistical analysis of species and regions will be presented.

Fig. 1 presents the comparison for the unrecoverable Maxwell Viscosity, η_m, which can be interpreted as the viscosity of the dashpot of a Maxwell element characterizing the long-time unrecoverable flow dissipation of energy. On average, η_m of all regions of the cow's and the pig's vitreous (except for the pig's central region) is about a factor of two greater than that of the human.

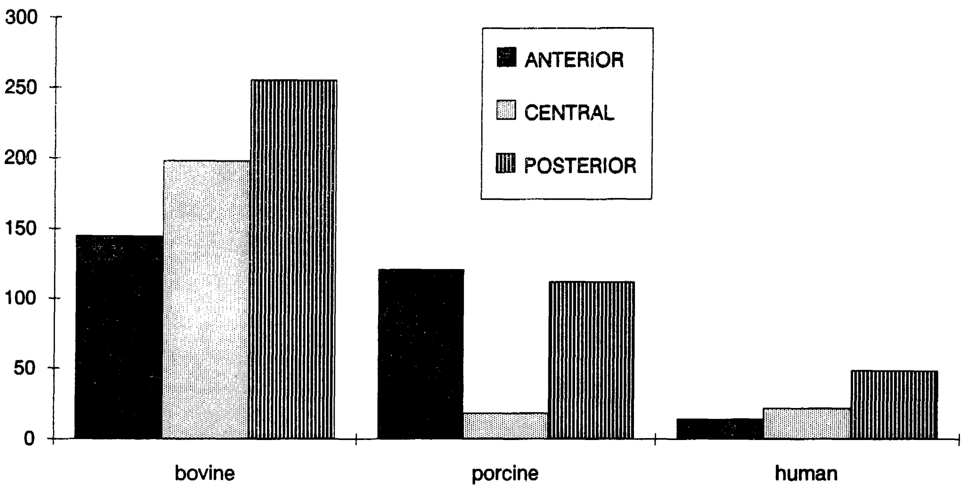

Fig. 1. Unrecoverable(Maxwell) Viscosity, η_m, as a function of species and region.

Table 2

Rheological Values for Porcine Vitreous as a Function of Region

	η_m	η_k	τ_m	τ_k	J_m	G_k
Anterior	121.18 +/- 34% (N=18)	8.28 +/- 32% (N=18)	9.65 +/- 34% (N=18)	0.84 +/- 27% (N=19)	0.08 +/- 75% (N=19)	8.01 +/- 21% (N=19)
Central	18.32 +/- 38% (N=15)	2.92 +/- 25% (N=15)	2.38 +/- 75% (N=16)	0.46 +/- 75% (N=17)	0.02 +/-160% (N=16)	10.40 +/- 27% (N=18)
Posterior	112.55 +/- 36% (N=19)	6.77 +/- 33% (N=19)	7.96 +/- 42% (N=20)	0.89 +/- 42% (N=20)	0.05 +/- 68% (N=20)	9.71 +/- 18% (N=20)

η_m of the pig's central region is significantly less than the two other regions, at least by a factor of nine. The mean value for the pig's central region is significantly less than that for the cow; however, no significant difference was detected between the pig's and the human's central vitreous regions. In

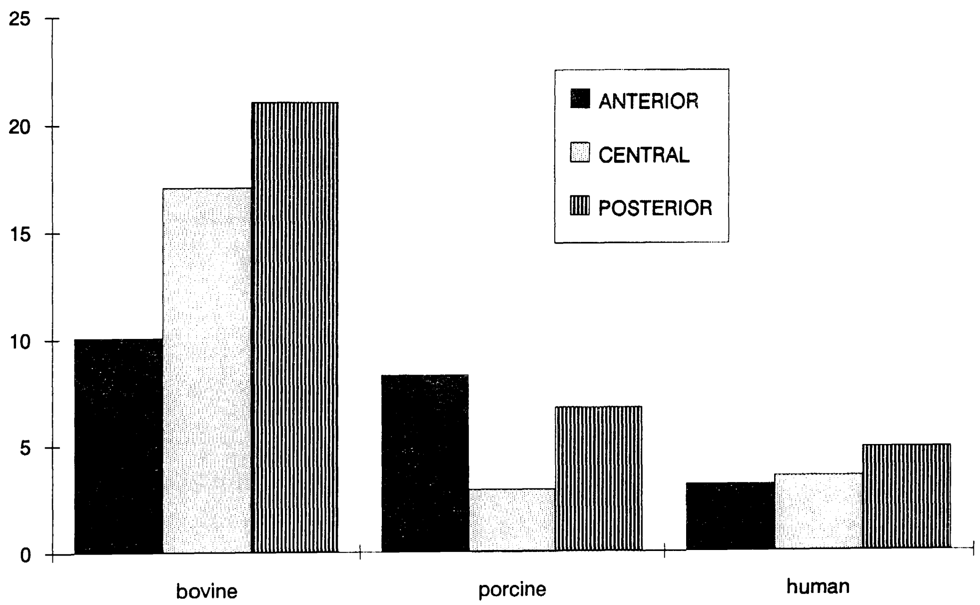

Fig. 2. Internal (Kelvin) viscosity, η_k, as a function of species and region.

particular, the data show that η_m of the central region in the cow is significantly greater than that of the central regions of both the human and the pig.

Comparison of values for the internal or Kelvin viscosity is given in Fig. 2. On the average, η_k of the pig and human central regions are significantly less than that of the cow. There is no significant difference in the mean value of η_k between the pig and the human central regions; however, η_k values for pigs and humans are significantly different in both the anterior and posterior regions. Values of the cow's vitreous from all three regions are significantly greater than those of the human.

From Figs. 1 and 2, it can be seen that the average η_m is at least 4–14 times greater than η_k in all regions of the vitreous of all three species. A greater η_m implies that either the true value is high or that there are many Kelvin elements with retardation times τ_k much greater than the time scale of the experiment (t_2). In such an instance, the retarded elastic element will appear as a flow, governed by the viscosity η_k (Alfrey, 1948). From the results of an investigation on η_m of a series of homologous polymers, Flory (1940) concluded that the Maxwell viscosity depends on the activation energy of segmental diffusion, as well as the degree of polymerization. On the other hand, the Kelvin viscosity is proportional only to the activation energy for the transport of segments of molecules, rather than the entire chain. Alfrey (1948) referred to the latter as the "internal friction" which will retard but not prevent the complete deformation recovery of a material; and it is probably due to the uncurling of long tangled polymer chains during the process of segmental diffusion of the

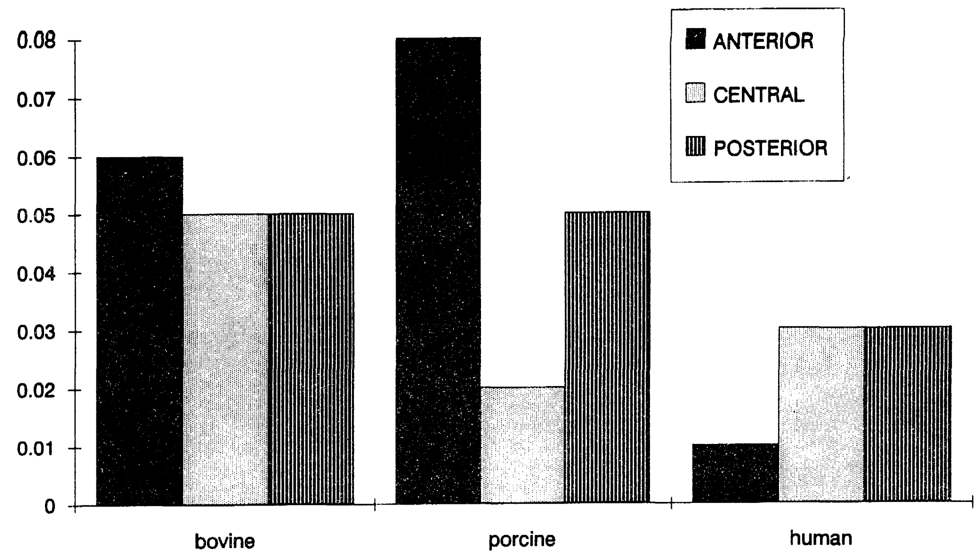

Fig. 3. Instantaneous elastic compliance J_m as a function of species and region.

chains. Therefore, the viscous term of the retarded elastic response is expected to have a much smaller absolute value at any temperature than the Maxwell viscosity. Our observations are in agreement with these theories.

The comparison of J_m values for the three species and regions is given in Fig. 3. The results of ANOVA indicate that there are no significant species differences in J_m for the posterior region. On the average, J_m in the anterior region is not significantly different between pig and cow; however, species differences were found to be significant between cow and human in both the anterior and central regions, and between pig and human in the anterior region. J_m of the pig's central region is significantly less than that of the cow, but not of the human.

As noted in our earlier paper (Lee *et al.*, 1992), values of the instantaneous compliance J_m should be interpreted with some caution, because the methodology cannot detect instantaneous jumps with frequency components greater than 30 Hz. For the vitreous samples whose values of J_m were not assumed to be zero (the 4-parameter model samples), no significant difference was observed, e.g., among the posterior vitreous regions of the three species or among the three regions of the cow's vitreous. Low average values of J_m may be due to a preponderance of samples with very high values of G_m, in which a better overall fit is obtained by arbitrarily setting J_m (= $1/G_m$) equal to zero. Thus while low individual values of J_m cannot be distinguished from zero, results of statistical comparisons may be considered valid, except that caution must be used with regard to interpretation of exact levels of significance.

The comparison of species and regions for G_k, the internal elastic modulus of the Kelvin spring, is given in Fig. 4. The G_k values of the pig's vitreous are about the same in all regions. The human samples show an increase in the anterior region values compared with the other two. In the bovine, there are significant differences between all three regions.

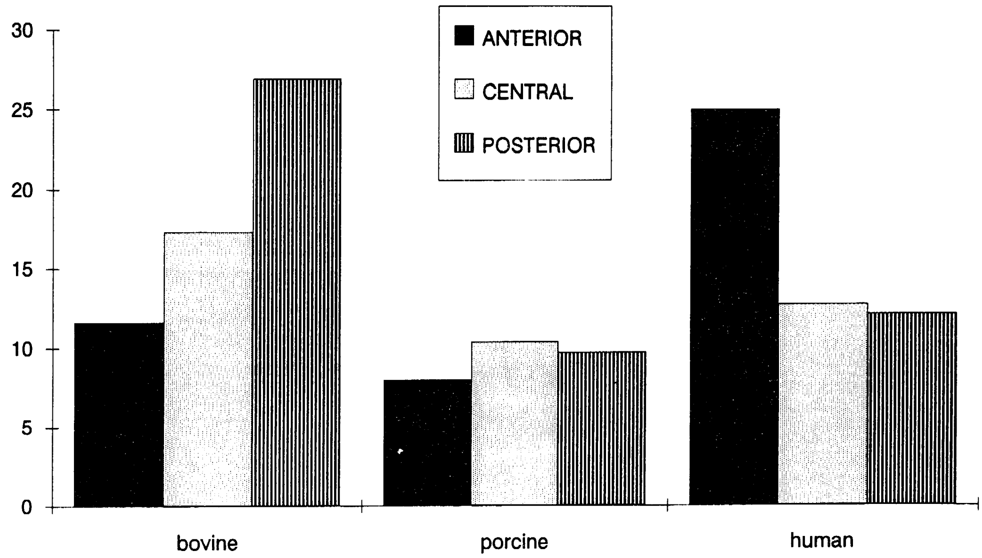

Fig. 4. Internal Elastic Modulus, G_k, as a function of species and region.

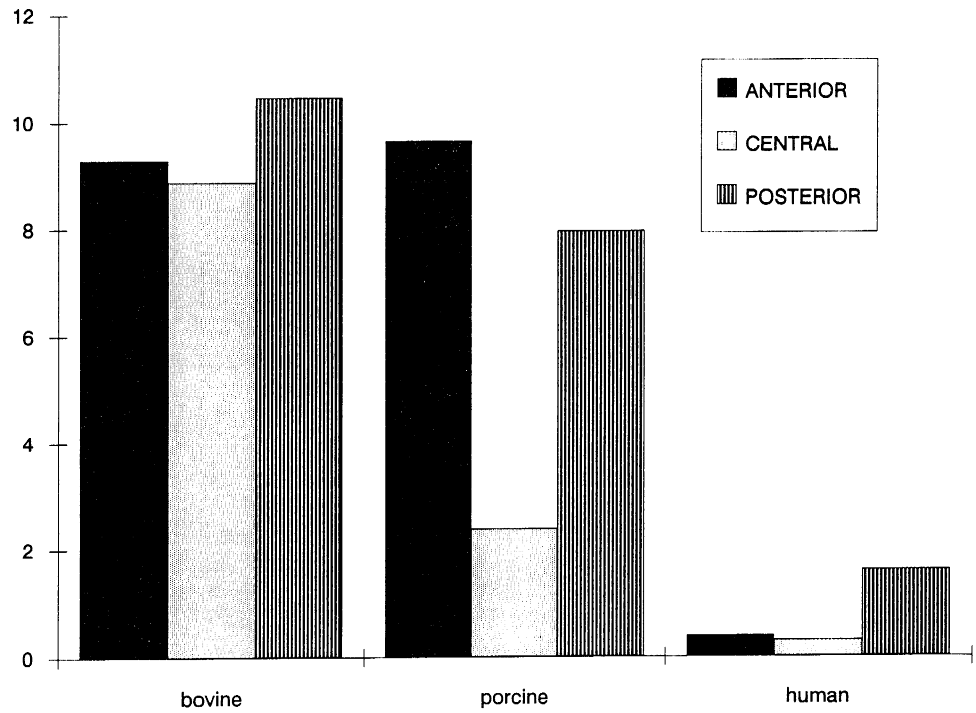

Fig. 5. Relaxation time, τ_m, as a function of species and region.

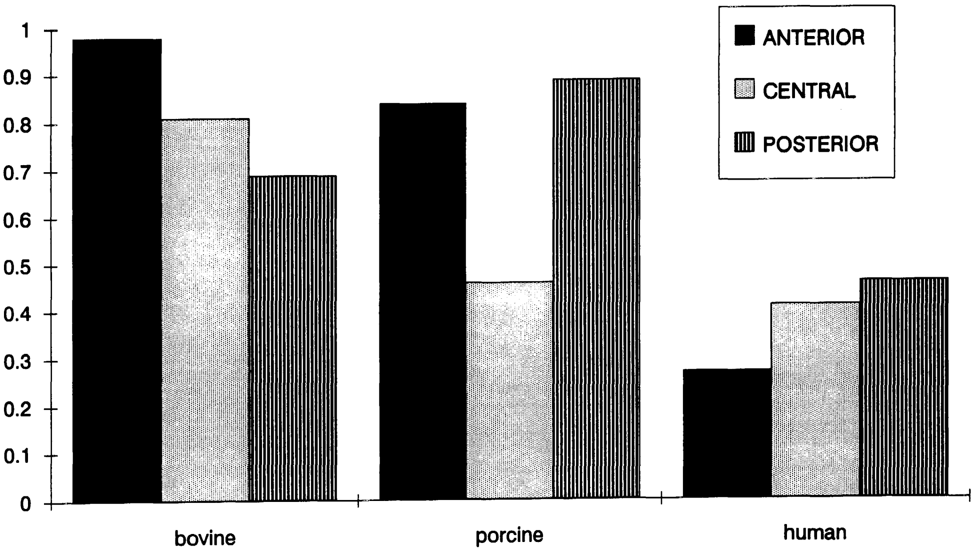

Fig. 6. Retardation time, τ_k, as a function of species and region.

Comparisons of the relaxation and retardation times, τ_m and τ_k , are given in Figs. 5 and 6, respectively. The average values of the relaxation time of samples from the human's anterior and central regions, and the pig's central region, are quite low. Because the value of τ_m is calculated from J_m for each sample (since $\tau_m = J_m\eta_m$), this result reflects the fact that J_m for many of these samples was not significantly different from zero, as indicated by the fact that the data were best fit by the three-parameter model. Therefore, for samples of low compliance, the relaxation time will be essentially correlated to the compliance.

In the porcine anterior and posterior regions, where values of the compliance J_m are high, t_m is higher and comparable to the values for the cow. There are no significant differences in these two regions between pig and cow. However, the relaxation time of the human posterior region is significantly less than that of the cow and pig. Since there is no difference in the mean values of J_m, a measure of elasticity, for the posterior region data of the three species, the differences in relaxation time observed must be due more to the significant differences in Maxwell viscosity η_m. Thus, the shorter relaxation time observed in the human vitreous samples is due not to a more dominant elastic (or energy storing) component in the human vitreous structure, namely G_m, but because the magnitude of the viscous (or energy dissipating) component of the human vitreous is much less than those of the two other species.

From Fig. 6, it is seen that on the average, τ_k values of the pig's anterior and posterior regions are significantly greater than that of the human, but there is no significant difference between central regions. In the cow, all three vitreous regions have a τ_k significantly greater than that of the human. No significant difference was found in τ_k between the anterior vitreous regions of the cow and the pig. There is a significant difference between these species in the posterior region.

Results of Statistical Analyses

Detailed statistical comparisons were carried out on the rheological parameters, to determine if significant differences exist between the different species and regions. Since there are three species and three regions, there are a number of different comparisons that can be carried out, considering regions and species separately or together. By including as many variables and data points in each analysis as possible, the analysis of variance techniques used can reveal the existence of significant pairwise and other differences despite the large variations in individual samples. Unless stated otherwise, all comparisons labeled significant are at the 95% confidence level. The major conclusions that may be drawn from these comparisons are as follows:

- An overall comparison, considering all parameters, species, and regions together, which maximizes the number of data points and variables in the analysis, shows that variations between all three species and all three regions of the vitreous are significant, *i.e.*, both species and region are significant variables. In addition, there is a significant interaction effect, indicating that when considered together, species and region display greater power to demonstrate differences than either variable alone. All the rheological variables, with the exception of G_k, exhibit this effect.
- Comparison of parameters between species, combining the data from all three regions into one data set, shows that there are significant differences in the rheological properties of the vitreous body among species, reflected in all six parameters.
- Comparison of parameters taking two species at a time, with all three regions combined into one data set, shows that there are significant differences in each paired comparison. For the pig and human, all six rheological parameters show these differences, for the pig and cow, all but J_m and τ_k are significantly different; and for the cow and human, all but G_k reflect significant species variations.
- Comparison of parameters in each region, taking species two at a time, shows that human vitreous is significantly different from cow in all regions; that there is no difference between human and pig central vitreous; and that pig and cow rheological properties differ only for the central region. Thus, the pig central region resembles human, while the remainder resembles cow.
- A comparison of parameters between regions with all three species combined, demonstrates that there are notable differences in the rheological properties among the three regions of the vitreous. Results of ANOVA suggest that this difference is due mainly to four of the six rheological parameters: η_m, η_k, τ_m, and J_m.
- Analysis of the regional variation within the data for each individual species shows that, for the human, there are significant differences for all six rheological parameters, although G_k and τ_k are significant only at the 94% level. For the pig, all parameters except for G_k show regional differences. For the cow, however, only the two viscosity parameters and the retarded elastic modulus G_k show significant regional differences.

Discussion

There are several possible factors that may determine the rheological behavior observed, such as differences in macromolecular and electrolyte concentrations. For example, the osmotic properties of HA have been shown to influence the size and shape of the molecules, and hence, viscosity values. It has been pointed out (Ogston, 1970) that connective tissue polysaccharides such as HA bind water. In solution, HA expands by a volume factor on the order of 1000. The larger physical domain occupied by these solvated molecules will result in a decrease of free volume and an increase of interparticle interactions. η_m is the factor that controls the unrecoverable Newtonian flow of a polymer system, and it is associated with the slippage of the polymer chains past one another (Alfrey and Gurnee, 1956). According to Flory's theory on the flow mechanism of polymer chains (Flory, 1940), viscosity is determined by the chain length and the frequency of segmental jump (the flow of a segment of a polymer chain). The chain length is proportional to the degree of polymerization. The segmental jump frequency is dependent on the temperature and the free volume in the system. Therefore, a decrease of free volume will definitely affect the viscosity of the overall system.

The retardation time τ_k of the Kelvin element is defined as the ratio of η_k to G_K. In a creep recovery experiment, when a load is suddenly removed from a sample, the strain will diminish. When a constant stress is removed from a Kelvin element, the displacement of the element will diminish exponentially as a function of time, and τ_k is the time that is required for the strain to fall to $1/e$ of the maximum displacement. Therefore, when the retardation time is short, strain recovers rapidly. The fact that retardation times of the cow and the pig vitreous (with the exception of the pig central region) are significantly greater than that of the human implies that when a load is suddenly removed from the vitreous, the displacement of the vitreous matrix will recover faster in the human than in either the pig's or the cow's vitreous.

The differences in the magnitudes of the two viscosity factors seem to correlate well with the differences observed in the gross physical state of the vitreous observed in this and the previous studies. Grossly, all regions of the cow vitreous, and the anterior and posterior regions of the pig vitreous, appeared to be a "thicker" gel, whereas the human gel more often presented a "thinner" consistency. Indeed, some of the anterior and the central human samples appeared "watery." Samples obtained from the pig central region also often presented a "thinner" consistency. Similarly, Balazs (1984) described cow vitreous as a "100% gel" in all regions but adult human, vitreous as only "40–80% gel." Eisner (1975) qualitatively confirmed in his postmortem studies of the dissected human vitreous that liquefaction begins in the central vitreous.

One possible explanation for some of these species differences is the probable differences in subject ages between the human and animal eyes. Whereas the porcine and bovine eyes are from relatively young animals, because of the nature of the collection, almost all human samples were obtained from elderly subjects. There is no apparent way to overcome this deficiency of the experimental protocol. However, age would not explain the striking similarity in rheological properties between pig central vitreous and human samples, which is highly significant and a major finding of this study. Another possible factor for the species differences is that either the properties or the organization of the vitreous structural elements are different in the different species. An investigation of such factors is possible, and will be discussed in future work.

The results suggest that in the vitreous, the level of "elasticity" does not seem to vary to a great extent among the three species that we studied. The difference in the "rigidity" or "thickness" of the vitreous gel observed in the different species seems to correlate well with the magnitude of the viscous components of the vitreous. This level of "elasticity" may be a prerequisite for the function of the vitreous in the eyes of all animal species. Varying the viscosity or the "thickness" of the vitreous gel may allow this connective tissue to perform its highly specialized functions in different animal species. One should also keep in mind that in the intact eye, vitreous function reflects the integration of mechanical properties of the various, somewhat arbitrarily designated regions that were sampled for analysis. Some other segmentation, which could vary with species, might be more revealing with regard to the relationship between rheology and function. Also, the relationship between rheology and function might depend on such anatomical and physiologic factors as upright *vs.* horizontal posture that would be reflected in different major optical axes.

From a physicochemical point of view, since samples that appear to be a "thicker gel" tend to have higher viscosities, one may conclude that the "thickness" of the vitreous gel is not a result of any permanent interparticle interaction, such as physical binding or covalent crosslinking between the vitreous structural elements, which would give the vitreous a more "elastic (or solid-like)" property. The apparent "thickness" of the vitreous gel may simply be because those samples are more "viscous." Such a behavior could be a result from a simple steric or close-packing interaction. The close packing interaction between molecules will decrease the free volume, and therefore increase the viscosity of the system. Such a system will show some resistance to flow (or deform) due to frictional force initially because the particles are closely packed together; however, any further deformational impact will result in a breakdown of the structure of the system. To investigate the role of the macromolecular elements in such behavior, a study was carried out to determine concentration levels of collagen and hyaluronic acid in the vitreous of the three species studied, as well as specific electrolyte levels that can affect their rheological behavior. The results of that study will be presented in the next paper of this series.

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